

Differential Calculus — Chain Rule

Chapter 2: Sine of Polynomial Function | Page 33

PROBLEM STATEMENT

Trigonometric Function of Polynomial

Find the derivative with respect to x of the function:

$$y = \sin(x^3 + x^2 + 8)$$

STEP 1 প্রদত্ত ত্রিকোণমিতিক বহুপদী ফাংশনটি উল্লেখ করি (Given Composite Function)

$$y = \sin(x^3 + x^2 + 8)$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [\sin(x^3 + x^2 + 8)]$$

STEP 3 সাইন ফাংশনের ওপর চেইন রুল প্রয়োগ করি ($\frac{d}{dx}[\sin u] = \cos u \frac{du}{dx}$)

$$\frac{dy}{dx} = \cos(x^3 + x^2 + 8) \cdot \frac{d}{dx}(x^3 + x^2 + 8)$$

STEP 4 অভ্যন্তরীণ ত্রিঘাত বহুপদী কোণের অন্তরক নির্ণয় করি (Differentiating Inner Polynomial Argument)

$$\frac{d}{dx}(x^3 + x^2 + 8) = 3x^2 + 2x + 0 = 3x^2 + 2x$$

STEP 5 মান বসিয়ে বন্ধনীযুক্ত বহুপদীটিকে সামনে লিখে চূড়ান্ত অন্তরজ পাই (Final Form)

$$\frac{dy}{dx} = (3x^2 + 2x) \cos(x^3 + x^2 + 8) = x(3x + 2) \cos(x^3 + x^2 + 8)$$

FINAL ANSWER
$$\frac{dy}{dx} = (3x^2 + 2x) \cos(x^3 + x^2 + 8)$$

Rigorous Insights & Mark Scheme

AMIE Section-A Standard Evaluation Protocol | Page 33, HW 09

✦ BOX 1: FORMULATION & CHAIN RULE ON POLYNOMIAL ARGUMENTS

- **General Formula:** $\frac{d}{dx} \sin(P(x)) = P'(x) \cos(P(x))$.
- For $P(x) = x^3 + x^2 + 8$, we have $P'(x) = 3x^2 + 2x$.
- Always enclose multi-term polynomials in parentheses: $(3x^2 + 2x) \cos(x^3 + x^2 + 8)$.

⚠ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Omitting Parentheses around $3x^2 + 2x$:** Writing $3x^2 + 2x \cos(x^3 + x^2 + 8)$ is a major mathematical syntax error (it means $3x^2 + [2x \cos(\dots)]$).
- **Differentiating the Constant:** Remembering that $\frac{d}{dx}(8) = 0$.

🌐 BOX 3: REAL DOMAIN & CRITICAL POINTS

- **Domain of Definition:** The function is defined and smooth on $\mathbb{R} = (-\infty, \infty)$.
- **Stationary Points:** $\frac{dy}{dx} = 0$ when:
 1. $3x^2 + 2x = 0 \implies x = 0$ or $x = -\frac{2}{3}$.
 2. $\cos(x^3 + x^2 + 8) = 0 \implies x^3 + x^2 + 8 = \frac{\pi}{2} + k\pi$.

💡 BOX 4: LEIBNIZ SUBSTITUTION (ALTERNATIVE VERIFICATION)

- Let $u = x^3 + x^2 + 8 \implies y = \sin u$.
- $\frac{dy}{du} = \cos u$.
- $\frac{du}{dx} = 3x^2 + 2x$.
- By Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \cos u \cdot (3x^2 + 2x) = (3x^2 + 2x) \cos(x^3 + x^2 + 8)$$

📊 BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 3 / 4 MARKS)

- **1.0 Mark:** Correct application of trigonometric derivative $\cos(x^3 + x^2 + 8)$.
- **1.0 Mark:** Accurate calculation of inner polynomial derivative $3x^2 + 2x$.
- **1.0 Mark:** Correct use of parentheses and finalized presentation $(3x^2 + 2x) \cos(x^3 + x^2 + 8)$.